NNN-C-00680A (DSA-DM)
May 24, 1972
Interim Revision of
NNN-C-680
September 22, 1961

INTERIM FEDERAL SPECIFICATION

CRUCIBLES, FILTERING AND IGNITION; AND

COVERS, LABORATORY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification NNN-C-680.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers porcelain ignition and filtering crucibles and crucible covers.

1.2 Classification.

1.2.1 Crucibles. Crucibles shall be of the following types and sizes, as specified:

Type I--Ignition crucibles.

Size 1	5 milliliters (ml.).
Size 2	10 ml.
Size 3	15 ml.

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Type I--Ignition crucibles (cont'd)

Size 4	30 ml. wide form
Size 5	50 ml.
Size 6	100 ml.
Size 7	250 ml.
Size 8	30 ml., high form
Size 9	150 ml.

Type II--Filtering crucibles.

Size 1	1.5 ml.
Size 2	13 ml.
Size 3	25 ml.
Size 4	40 ml.

1.2.2 Crucible covers. Crucible covers shall be of the following sizes, as specified:

Size 1	17 mm.
Size 2	26 mm.
Size 3	34 mm.
Size 4	38 mm.
Size 5	44 mm.
Size 6	52 mm.
Size 7	66 mm.
Size 8	77 mm.
Size 9	105 mm.
Size 10	45 mm.
Size 11	90 mm.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.

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SPECIFICATIONS (Cont'd.)

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Porcelain. Porcelain shall be suitable for the purpose intended and shall pass the tests specified herein.

3.1.1.1 Resistance to thermal shock. Crucibles and crucible covers shall show no evidence of cracking or crazing when tested as specified in 4.4.1.

3.1.2 Glaze. The glaze shall offer a high degree of inertness and have a thermal coefficient of linear expansion approximately equal to that of the porcelain body. Crucibles shall be glazed inside and outside with the exception of the outside bottom surface. Covers shall be glazed inside and outside with the exception of the bottom edge.

3.1.2.1 Glaze adhesion. Permanent adhesion of the glaze to another glazed porcelain surface of the same material shall not occur below 1150° C. when tested as specified in 4.4.2.

3.1.2.2 Permanent adhesion. Adhesion shall be considered permanent when, after cooling to room temperature, the touching surfaces are annealed as one, or the touching surfaces separate with a portion of one surface adhering to and becoming part of the other surface.

3.1.2.3 Solubility. No watermark shall remain on the inner surface of the crucible and no pink color shall develop in the test solution when tested as specified in 4.4.3.

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3.1.2.4 Resistance to chemical attack. Resistance to chemical attack shall be measured by loss of weight by boiling in a solution of sodium hydroxide. Weight loss shall be not more than 0.144 percent when tested as specified in 4.4.4.

3.2 Design.

3.2.1 Shape.

3.2.1.1 Crucibles. Type I crucibles shall have a flat non-perforated base suitable for resting firmly on a flat surface. The side wall shall be formed by a sloping, convex surface extending upward from the base to the rim. Type II crucibles shall have the same style and design as type I except that the base shall be perforated.

3.2.1.2 Crucible covers. The crucible covers shall be designed for use with the respective sizes of crucibles (see tables I and II). There shall be a ring molded in the center of the top of the crucible covers perpendicularly to serve as a handle.

3.2.2 Dimensions and capacities.

3.2.2.1 Crucibles, type I. The dimensions and capacities shall be as shown in table I.

3.2.2.2 Crucibles, type II. The dimensions and capacities shall be as shown in table II.

3.2.2.3 Crucible covers. The dimensions shall be as shown in tables I and II. In addition, all crucible covers, of a particular size, shall fit all crucibles of the appropriate size without binding when tested as specified in 4.4.5.

Table I Dimensions of type I crucibles and crucible covers

CRUCIBLES					COVERS	
Size	Capacity ml. *	Top outside diameter mm. **	Base outside diameter mm. **	Height mm. **	Inside diameter mm. **	Size
1	5	24	12	20	26	2
2	10	30	15	26	34	3
3	15	35	18	29	38	4
4	30	50	18	31	52	6
5	50	58	22	36	66	7
6	100	73	27	45	77	8
7	250	99	37	61	105	9
8	30	43	--	37	45	10
9	150	84	--	52	90	11

* Tolerance ± 15 percent
** Tolerance ± 1 mm.

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Table II Dimensions of type II crucibles and crucible covers

CRUCIBLES					PERFORATIONS		COVERS	
Size	Capacity ml. *	Height mm. **	Top outside diameter mm. **	Base outside diameter mm. **	Each diameter mm. **	Each diameter mm. **	Inside diameter mm. **	Size
1	1.5	15	15	12	0.5	7.5	17	1
2	13	34	29	18	0.7	11.0	34	3
3	25	41	36	22	0.7	16.0	38	4
4	40	47	40	25	0.6	19.0	44	5

* Tolerance ± 15 percent ** Tolerance ± 1 mm. *** Tolerance ± 0.02 mm.

3.3 Identification marking. Crucibles and crucible covers shall be permanently and legibly marked with the name or trademark of the manufacturer. Markings shall be legible after shock, adhesion, and chemical tests have been performed.

3.4 Workmanship. Crucibles and crucible covers shall be symmetrical in shape, uniform in size, and free from cracks, crazing, and other irregularities or foreign matter which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table III. Unit of product for examination purposes shall be the crucible or crucible cover, as specified.

Table III Sampling for examination

Type defect	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product for the purposes of testing shall be the crucible or crucible cover, as specified (see 6.2). The inspection level shall be as indicated in table IV, and the acceptance number shall be zero.

Table IV Test procedure

Characteristic	Requirement	Test procedure	Inspection
Thermal shock resistance	3.1.1.1	4.4.1	S-2
Glaze adhesion	3.1.2.1	4.4.2	S-2
Solubility	3.1.2.3	4.4.3	S-2
Resistance to chemical attack	3.1.2.4	4.4.4	S-2
Binding	3.2.2.4	4.4.5	S-2

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4.3 Examination. The crucibles or the crucible covers, as specified (see 6.2), shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with tables III and IV. The crucibles and crucible covers shall be examined for defects including, but not limited to, those listed in table V.

Table V Classification of defects

Categories	
101	Capacity and dimensions incorrect.
102	Crucibles and crucible covers not free from cracks, crazing, chipping, pinholes, blistering, crawling, eggshelling, orange peel, or shivering.
103	Crucibles and crucible covers not uniformly glazed.
104	Crucibles and crucible covers not free from grease, oil, or foreign matter.
Minor	
201	Identification marking missing, incorrect, or illegible.

4.4 Tests. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.1 Thermal shock resistance. The crucibles or crucible covers shall be heated to $160^{\circ} \pm 2^{\circ}$C. in an oven for 30 minutes, removed, and immediately plunged into water at a temperature of $9^{\circ} \pm 1^{\circ}$C.

4.4.2 Glaze adhesion. Place two crucibles or crucible covers so that the glazed surfaces are in contact with each other and heated to 1150°C. in an electric muffle-type furnace for 30 minutes. Allow to cool by radiation to room temperature.

4.4.3 Solubility. Crucibles or crucible covers, taken for test, shall be cleaned and rinsed three times with distilled water, using a new quantity of distilled water for each rinsing. The crucibles or crucible covers shall be air-dried in an inverted position for one minute after rinsing. After the final drying operation, each type I crucible shall be half-filled with distilled water at 60°C. containing 0.2 percent phenolphthalein solution (prepared from 1 gram of phenolphthalein in 100 ml. alcohol) and autoclaved under 15 pounds steam pressure for 30 minutes. Crucible covers and type II crucibles shall be immersed individually in distilled water containing 0.2

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percent phenolphthalein solution in a beaker of suitable size, the crucible covers upside down and the type II crucibles in an upright position. The distilled water used in this test shall have just been boiled for 10 minutes and allowed to cool to 60° C., prior to the addition of the phenolphthalein solution.

Note: Glassware used in this test shall not affect test solution.

4.4.4 Resistance to chemical attack. Tared crucibles or crucible covers shall be boiled in 10 percent solution of sodium hydroxide for 8 hours, rinsed, dried and reweighed. Weight loss shall be calculated as follows:

$$L = \frac{100\ (B-A)}{B}$$

Where: L = percent weight loss
B = weight before boiling in grams
A = weight after boiling in grams

4.4.5 Binding. Crucible covers, in the sample, shall be rotated 360° in place, on accepted crucibles of appropriate size.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.2 Level A.

5.1.2.1 Unit package.

5.1.2.1.1 Crucibles. Crucibles shall be wrapped in kraft paper. Wrapping shall be secured with tape.

5.1.2.1.2 Crucible covers. Crucible covers shall be wrapped in kraft paper and secured as specified in 5.1.2.1.1.

5.1.2.2 Intermediate package. Twelve crucibles or twelve crucible covers of the same size packaged in accordance with 5.1.2.1 shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. The box shall be provided with full height partitions forming a cell for each

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item or contain sufficient cushioning material. Closure of box shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Five hundred seventy six (576) units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Five hundred seventy (576) units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

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5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated on water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I crucibles are intended for use in ignition of chemical substances for ash content.

6.1.2 Type II. Type II crucibles are intended for use in filtration.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal Stock Number (FSN).
c. Type and size of crucible required.
d. Size of crucible cover required.
e. Level of packaging and packing required (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
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